Maven

What is Maven?

- Apache Maven is a powerful build automation and project management tool primarily used for Java-based projects. It simplifies the software development process by handling:
  - Dependency management
  - Build processes (compile, test, package, deploy)
  - Project documentation
  - Reporting

POM (Project Object Model)

- The heart of Maven - an XML file (pom.xml)
- Contains project configuration, dependencies, and build settings

```
<project>
  <modelVersion>4.0.0</modelVersion>
    <!-- Project coordinates (like an address) -->
  <groupId>com.mycompany</groupId>
  <artifactId>my-app</artifactId>
  <version>1.0.0</version>
    <!-- Dependencies -->
  <dependencies>
    <dependency>
      <groupId>junit</groupId>
      <artifactId>junit</artifactId>
      <version>4.13.2</version>
      <scope>test</scope>
    </dependency>
  </dependencies>
</project>
```

Dependencies & Repositories

- Central Repository
  - Maven's default public repository (contains millions of libraries)

- Local Repository
  - ~/.m2/repository on your machine (caches downloaded dependencies)

- Private Repositories
  - Companies often set up their own (Nexus, Artifactory)

Build Lifecycle & Phases

- Default - compile, test, package, install, deploy
  - Project deployment

- Clean – clean
  - Clean previous build artifacts

- Site – site
  - Generate project documentation

Common Commands

- mvn compile # Compiles source code
- mvn test # Runs unit tests
- mvn package # Creates JAR/WAR file
- mvn install # Installs to local repository
- mvn clean # Deletes target directory

Why Use Maven?

- Automatic Dependency Management
  - Downloads required libraries automatically
  - Handles transitive dependencies (libraries your dependencies need)
- Standardized Project Structure
  - Consistent across all Maven projects
  - Easy for new developers to understand
- Large Ecosystem
  - Thousands of plugins for various tasks
  - Integrates with CI/CD tools (Jenkins, GitHub Actions)
- Declarative Configuration
  - Define WHAT you want, not HOW to do it
- Built-in Best Practices
  - Standardized testing, packaging, deployment

Quick Start Example

```
mvn archetype:generate -DgroupId=com.example \
                -DartifactId=my-app \
                -DarchetypeArtifactId=maven-archetype-quickstart \
                -DinteractiveMode=false
```